FOOTBALL ANALYTICS · LONG-WEEKEND PROJECT

I graded 322,169 football decisions.

Not whether the pass arrived.
Whether it was the **right** pass.

Swipe →

THE PROBLEM

Every passing stat grades the outcome.

Completion % → did it arrive?

Progressive passes → did it go forward?

Expected assists → did it lead to a shot?

None of them ask: **was it the best option he had?**

Scouts rate "decision-making" constantly. Almost nobody measures it.

HOW IT WORKS

Decision Value Added (DVA)

1

Build the menu

From StatsBomb 360 freeze frames: every visible teammate to feet, every pass into a run, carrying, and shooting.

2

Value every option

How much each one raises or lowers the team's expected goals over the next 10 actions, scored by machine-learning models.

3

Compare to a typical player

A choice model learns what an average player picks from the same menu. DVA = value he chose minus value a typical player would choose.

SEEING THE RUNS

Freeze frames are snapshots. Players move.

By matching every player between consecutive frames, the model estimates who is running where. A moving teammate gets a second option: the spot where the ball would meet his run.

1 in 5

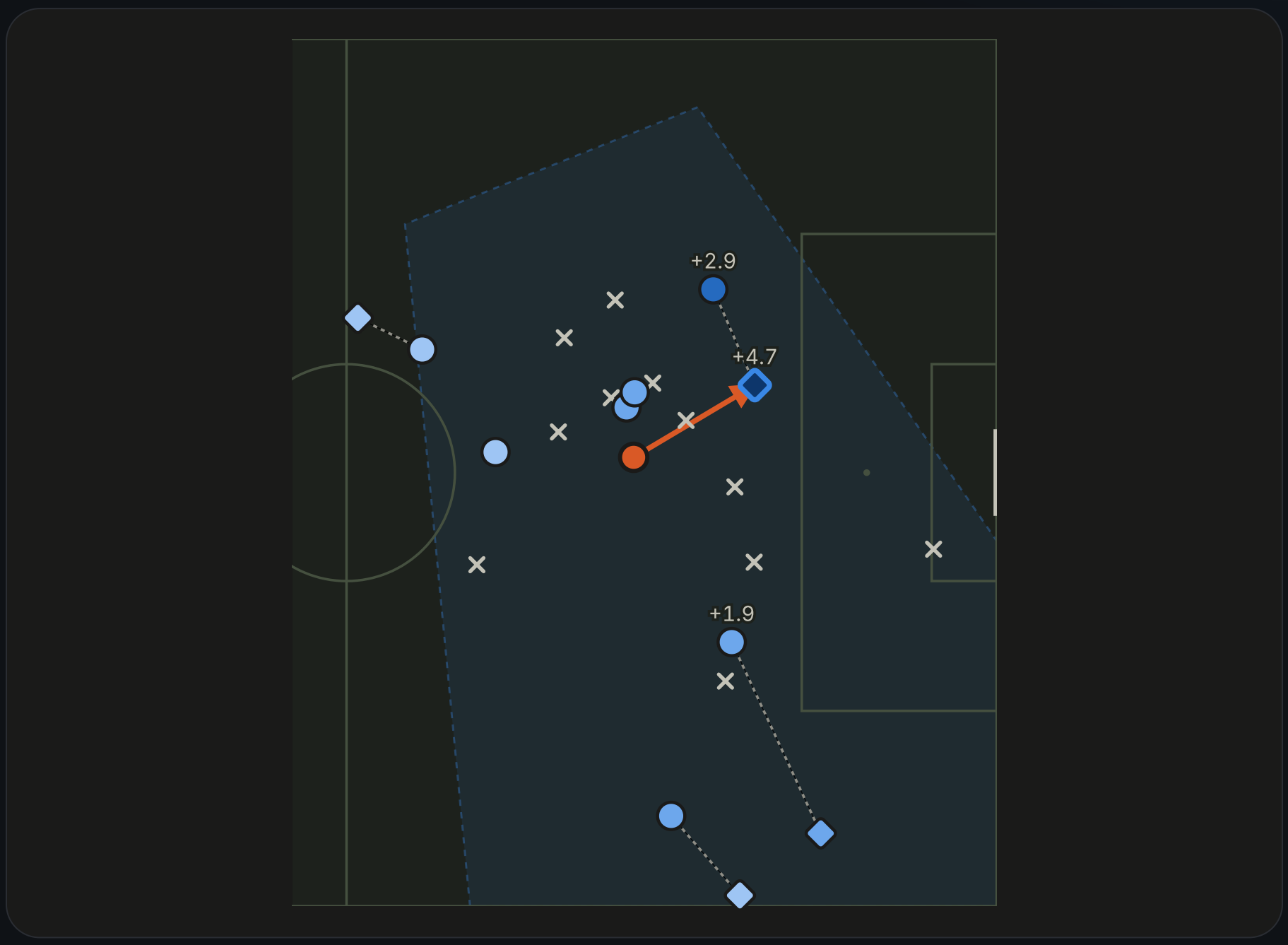
passes is played into a teammate's run. Without this, those decisions are invisible.

807K

"pass into run" options valued.

WHAT THE MODEL SEES

Messi, El Clásico, minute 91



Pass played · To feet · Into run · Opponent

Most options are safe balls to feet. The best is a teammate running toward the box. Messi plays it into his path: **+4.7** xG per 100 passes vs **+1.9** for a typical pick.

THE SCALE

Built on every men's competition with public 360 data

322K
graded decisions

293
matches

3.4M
options valued

8
competitions

Euro 2020 · World Cup 2022 · Euro 2024
Barcelona 20/21 · PSG 21/22 & 22/23
Inter Miami 2023 · Leverkusen 23/24

IS IT REAL?

Decision-making shows up as a repeatable trait.

0.64

reliability. Split each player's matches in half: his rating in one half predicts the other. Pure noise would score 0.

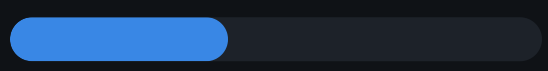
0.97

rank agreement when re-rated using only options inside the camera view, so unseen defenders aren't driving it.

Every model is tested on matches it never saw.

THE LEADERBOARD

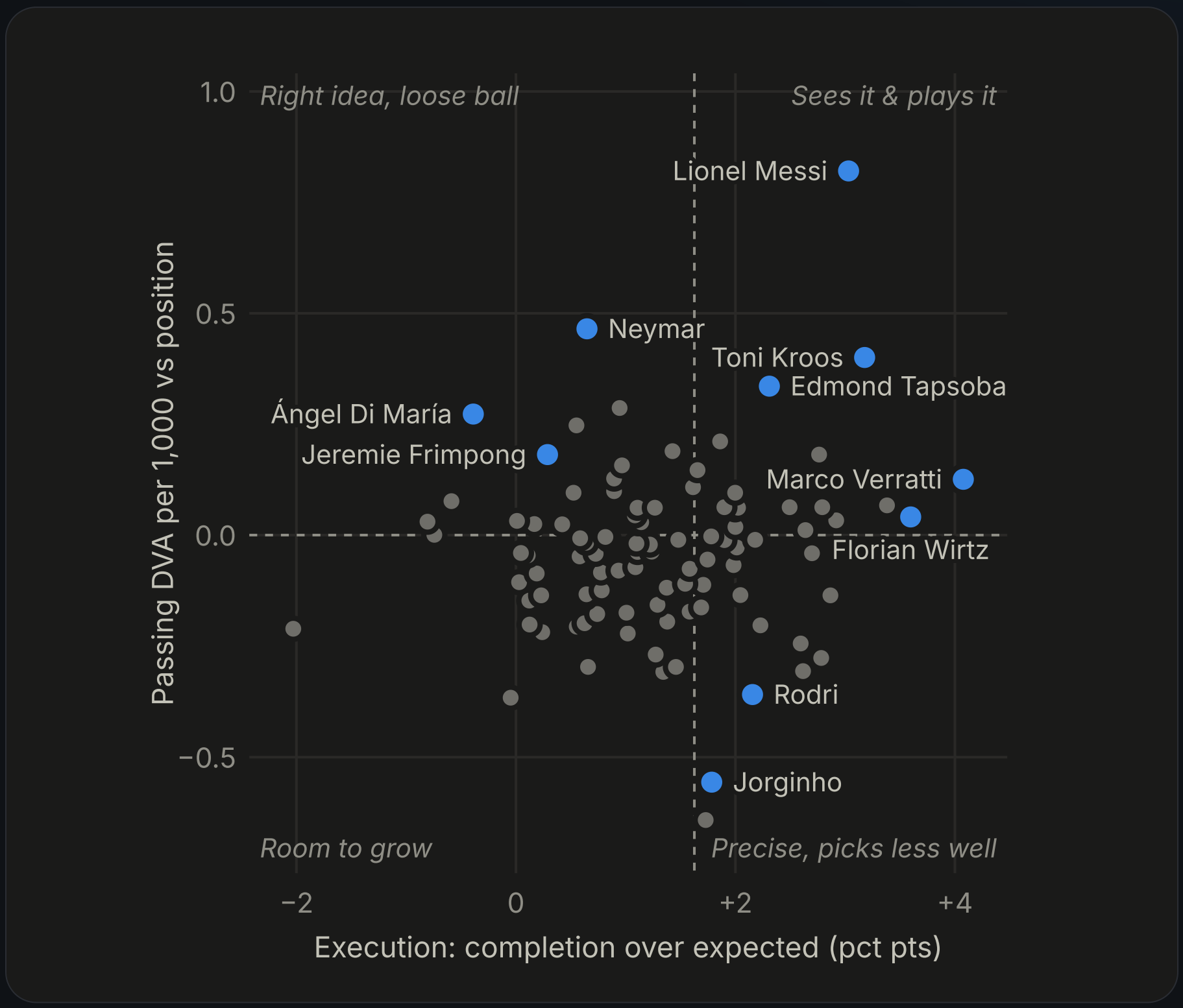
Best passing decisions (vs own position)

1	**Lionel Messi** Paris Saint-Germain · Forwards		+0.82
2	**Neymar** Paris Saint-Germain · Attacking mid & wide		+0.47
3	**Toni Kroos** Germany · Central midfield		+0.40
4	**Edmond Tapsoba** Bayer Leverkusen · Centre-backs		+0.34
5	**Ricardo Rodríguez** Switzerland · Centre-backs		+0.29
6	**Ángel Di María** Paris Saint-Germain · Attacking mid & wide		+0.27
7	**Pierre-Emile Højbjerg** Denmark · Central midfield		+0.25
8	**Odilon Kossonou** Bayer Leverkusen · Centre-backs		+0.21
9	**Sergio Ramos** Paris Saint-Germain · Centre-backs		+0.19
10	**Jeremie Frimpong** Bayer Leverkusen · Full-backs		+0.18

xG added per 1,000 passes vs a typical player in the same role. Min. 400 passes.

DECIDING ≠ EXECUTING

Picking the right pass and playing it well are different skills.



Choosing ambitious targets lowers completion %, which makes completion % a poor proxy for good decisions.

WHAT DIDN'T WORK (YET)

A model you can trust tells you where it breaks.

Pass, carry or shoot?

Choosing between actions isn't measurable as a skill yet (reliability 0.26). No ratings published.

Does it survive a transfer?

Only 20 players changed team in the public data. Too few to say either way.

The camera only sees part of the pitch

Stress-tested and not driving the ratings, but far-side switches can still be missed.

WHAT'S NEXT

More players who've changed teams, tracking data, and finer roles.

If you work in football analytics or recruitment:
what would make this genuinely useful in a scouting room?

Let me know in the comments.

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Data: StatsBomb Open Data (360 freeze frames).